ASPECT VISION

ब्रह्मास्त्र

BEU PATNA

2nd SEMESTER

ENGINEERING GRAPHICS & DESIGN

VVI QUESTIONS

WITH SOLUTION

2024

सभी QUESTIONS यहीं से आएंगे

Curated by expert's.

Q1. On a road map, a scale of miles is shown. On measuring from this scale, a distance of 25 miles is shown by a line 10 Cm. long. Construct this scale to read miles and to measure upto 40 miles. Construct a comparative scale, attached to this scale, to read kilometres upto 60 kilometres. 1 mile = 1.609.

(i) Scale of miles:

Length of the scale= $\frac{10 \times 40}{25} = 16Cm$

Draw a line 16 cm long and construct plain scale to show miles.

(ii) Scale of Kilometres:

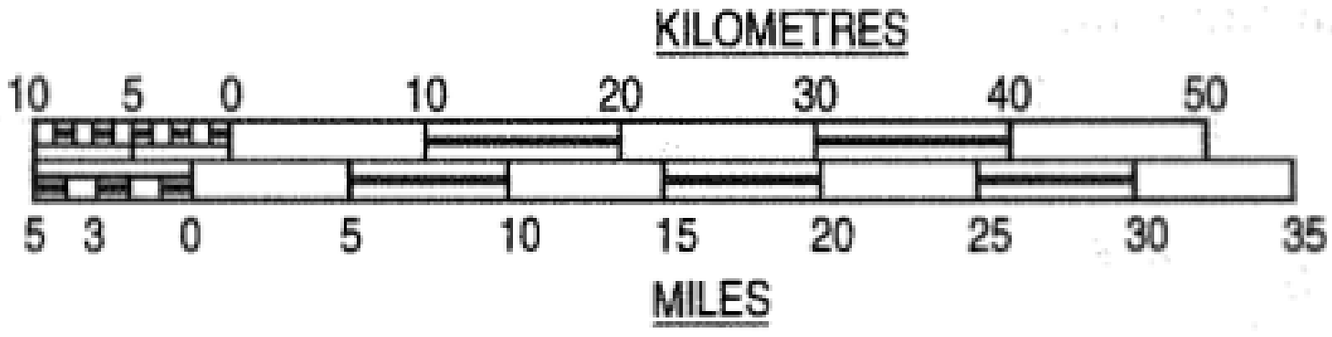
$$RF = \frac{10}{25 \times 1.609 \times 1000 \times 100} = \frac{1}{402250}$$

Length of the scale = $\frac{1}{402250} \times 60 \times 1000 \times 100 = 14.9Cm$

Draw a parallel 14.9 cm line below the miles scale, divide it into 60 equal parts, and label every 10th division as 10, 20, 30, etc.

Connect Both Scales:

Label the top scale "Miles" and the bottom scale "Kilometres."



Q2. Construct an ellipse when the distance of the focus from directrix is equal to 50 mm. and eccentricity is 2/3

(i) Draw any vertical line AB as directrix.

(ii) At any point C on it, draw the axis perpendicular to the AB (directrix).

(iii) Mark a focus F on the axis such that CF = 50 mm.

(iv) Divide CF into 5 equal divisions (sum of numerator and denometer of the eccentricity.).

(v) Mark the vertex V on the third division-point from C

Thus, eccentricity, e = (VF)/(VC) = 2/3

(vi) A scale may now be on the axis (as explained below), which will directly give the distances in the required ratio.

(vii) At V, draw a perpendicular VE equal to VF. Draw a line joining C and E

Thus, in triangle CVE, VE/(VC) = VF/(VC) = 2/3

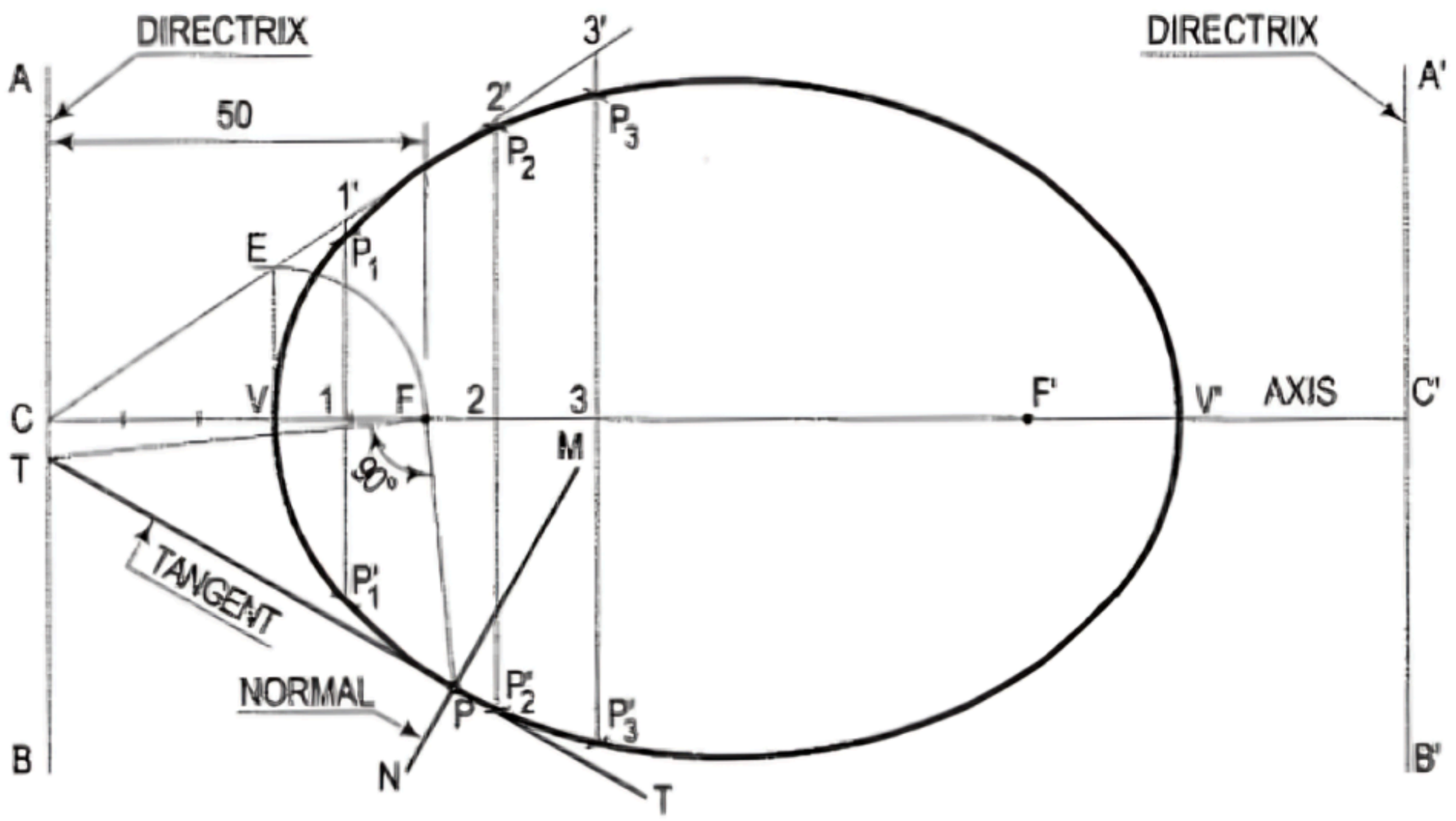
(viii) Mark any point 1 on the axis and through it, draw a perpendicular to meet CF-produced at 1'.
(ix) With centre F and radius equal to 1-1', draw arcs to intersect the perpendicular through 1 at points P1 and P' 1

These are the points on the ellipse, because the distance of p1 from AB is equal to C1

P1F=1-1'

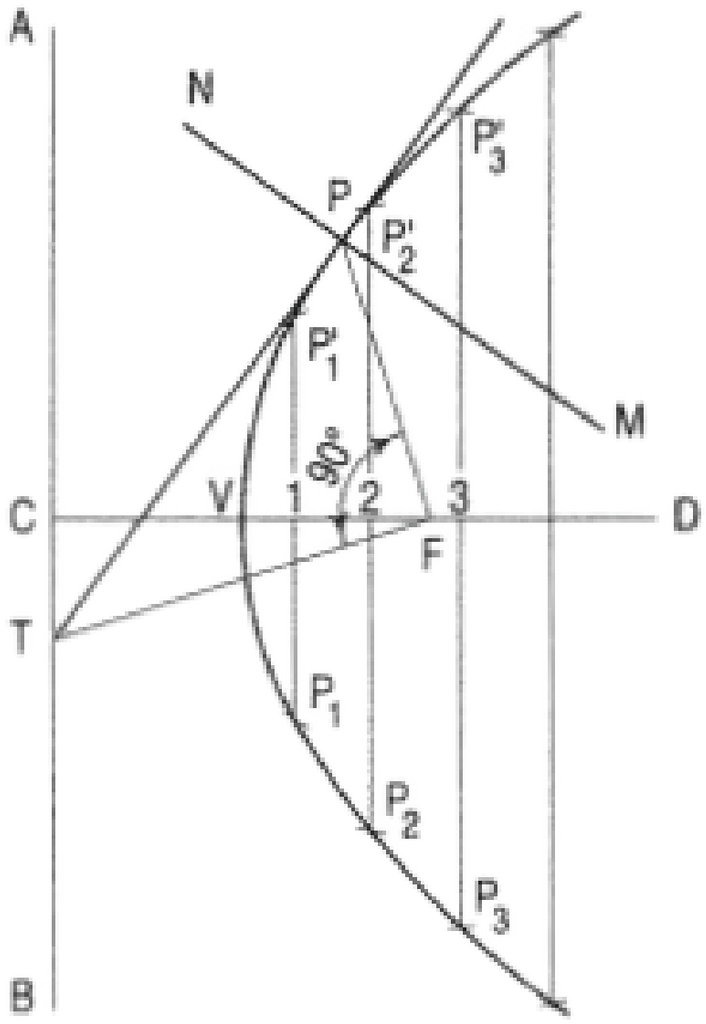
and $$\frac{1-1'}{C1} = \frac{VF}{FC} = \frac{2}{3}$$

Similarly, mark points 2, 3 etc. on the axis and obtain points p2, p'2, p3 and p'3 etc.
(x) Draw the ellipse through these points. It is a closed curve having two foci and two directrices.



Q2. Construct an parabola when the distance of the focus from directrix is equal to 50 mm.

(i) Draw the directrix *AB* and the axis *CD*.
(ii) Mark focus *F* on *CD*, 50 mm from *C*.
(iii) Bisect *CF* in *V* the vertex (because eccentricity = 1).
(iv) Mark a number of points 1, 2, 3 etc. on the axis and through them, draw perpendiculars to it.
(v) With centre *F* and radius equal to C1, draw arcs cutting the perpendicular through 1 at P_1 and P'_1.
(vi) Similarly, locate points P_2 and P'_2, P_3 and P'_3 etc. on both the sides of the axis.
(vii) Draw a smooth curve through these points. This curve is the required parabola. It is an open curve.



Q3. A thin circular disc of 50 mm diameter is allowed to roll without slipping from upper edge of sloping plank which is inclined at 15 degree with the horizontal plane. Draw the curve traced by the point on the circumference of the disc.

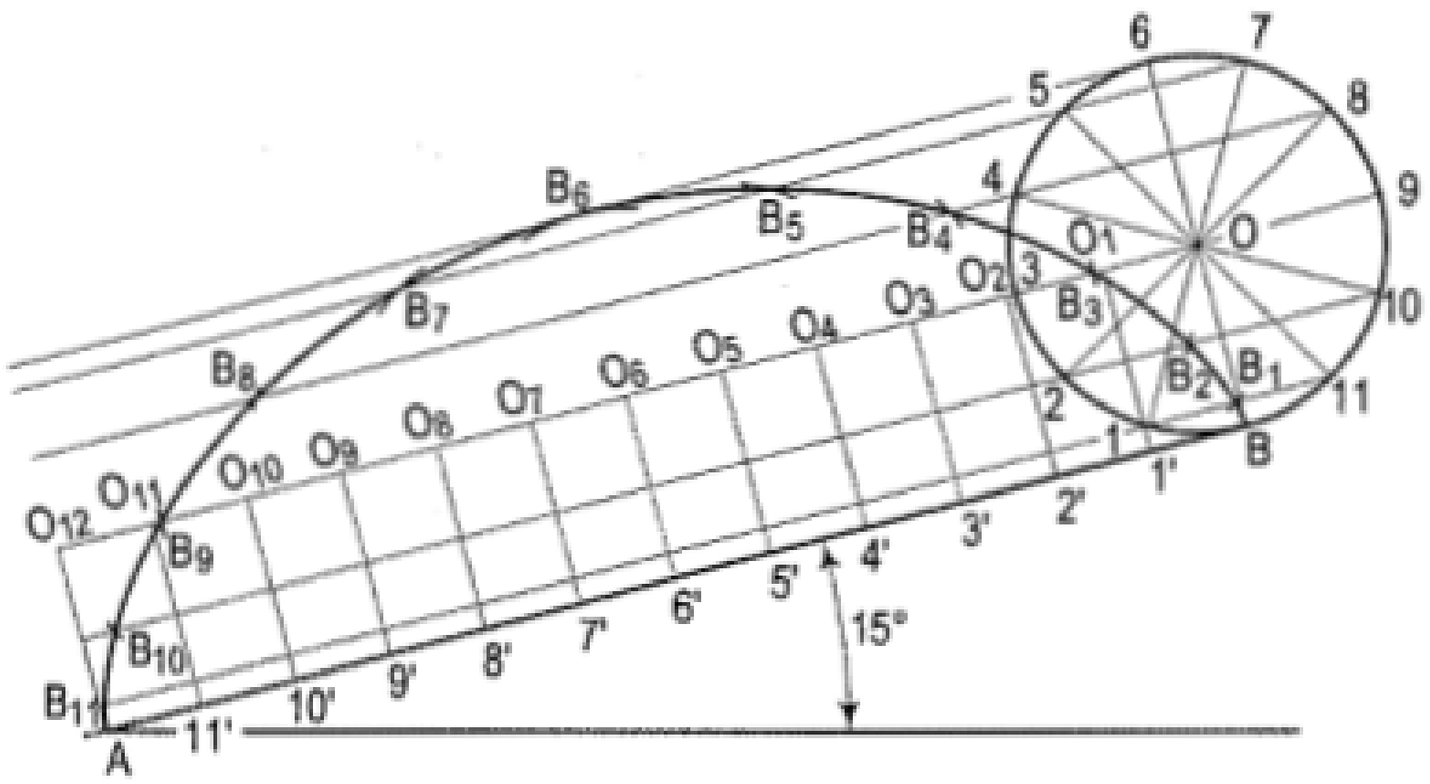
(i) Draw a line AB of length 157 mm (i.e. πD) at angle of 15° with the horizontal.

(ii) Draw a circle of 50 mm diameter at upper edge B as shown and divide the circumference of the circle in twelve equal parts. Name 1, 2, 3, to 11. Draw parallel lines from 1, 2, 3, 4 to 11 to the AB. Take B as generating point on the circumference.

(iii) Draw a parallel line from the centre of circle O to line AB and equal to length of AB. Divide this line in 12 parts. Name O_1 to O_{12} as shown.

(iv) As the disc rolls from position B, the centre of disc simultaneously moves from O to O_1. Draw an arc with radius as 25 mm and centre as O_1 to intersect the line drawn from the point 1 parallel to AB. Mark intersection point as B_1. This point is on the cyloid.

(v) Similarly draw arc from the points O_2 to O_{12} such that it intersects each line drawn from 2 to 11 respectively. Join the points by smooth curve. This curve is the cycloid.



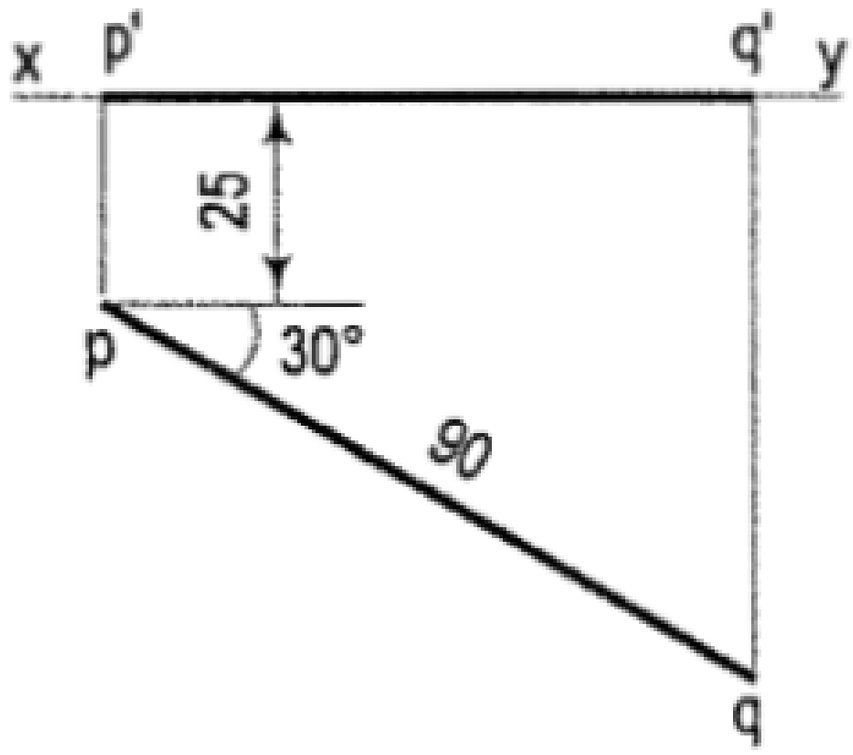
Q1. A line PQ 90 mm long, is in the HP and makes an angle of 30 degree with the VP its end P is 25 mm in front of the VP. Draw its projections.

As the line is in the H.P., its top view will show the true length and the true inclination with the V.P. Its front view will be in *xy*.

(i) Mark a point *p*, the top view 25 mm below *xy*. Draw a line *pq* equal to 90 mm and inclined at 30° to *xy*.

(ii) Project *p* to *p'* and *q* to *q'* on *xy*.

pq and *p'q'* are the required top view and front view respectively.



Q2. A line AB, 80 mm long, makes an angle of 60° with HP and lies in an auxiliary vertical plane (AVP), which makes an angle of 45° with the VP. Its end A is 10 mm away from both the HP and VP. Draw the projections of AB and determine (a) its true inclination with the VP and (b) its traces.

please see in the video lecture

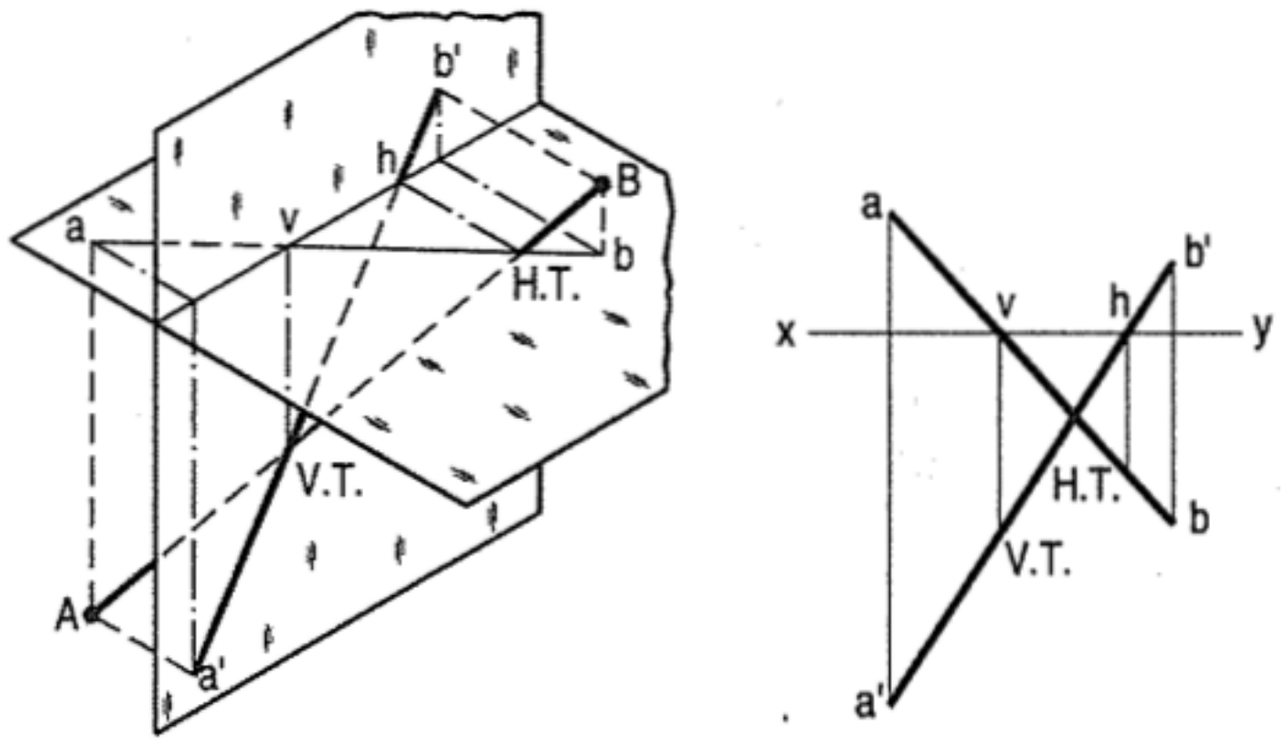
Q3. A point A is 50 mm below the H.P. and 12 mm behind the V.P. A point B is 10 mm above the H.P. and 25 mm in front of the V.P. The distance between the projectors of A and B is 40 mm. Determine the traces of the line joining A and B.

Draw the projections *ab* and *a'b'* of the line *AB*.

Method I: (fig. 10-29)**:**

(i) Through *v*, the point of intersection between *ab* and *xy*, draw a projector to meet *a'b'* at the V.T. of the line.

(ii) Similarly, through *h*, the point of intersection between *a'b'* and *xy*, draw a projector to cut *ab* at the H.T. of the line.



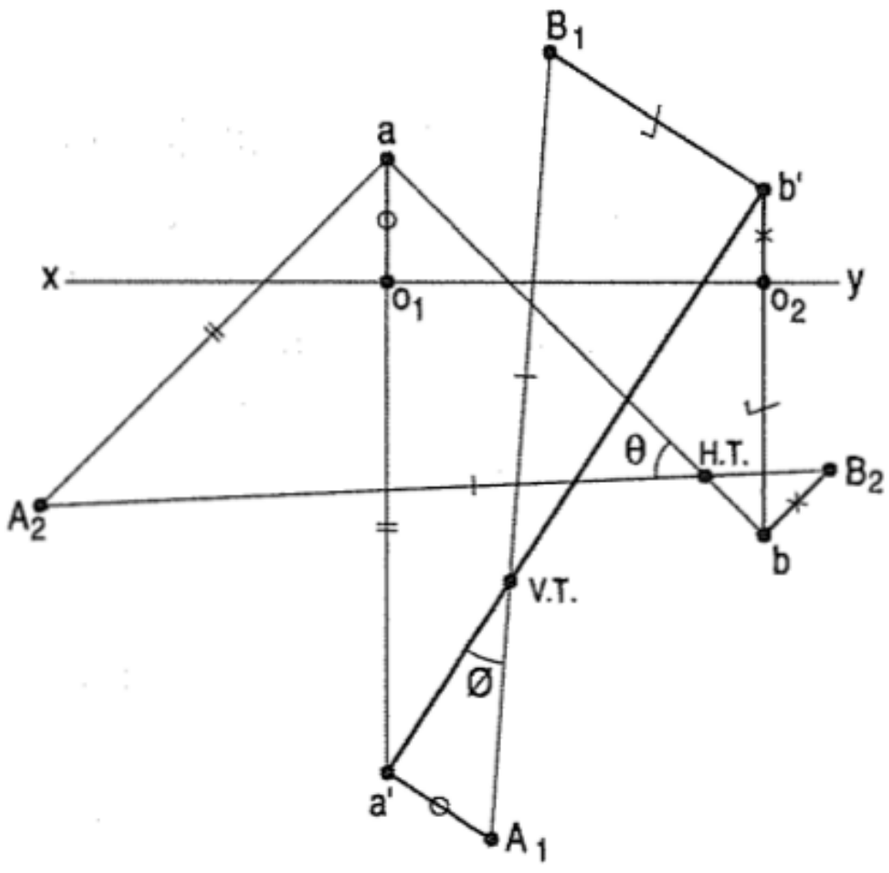
Method II: (fig. 10-30)**:**

At the ends *a'* and *b'*, draw perpendiculars to *a'b'*, viz. $a'A_1$ equal to ao_1 and $b'B_1$ equal to bo_2 on its opposite sides (as *a* and *b* are on opposite sides of *xy*).

Draw the line A_1B_1 intersecting *a'b'* at the V.T. of the line.

Similarly, at the ends a and b, draw perpendiculars to ab, viz. aA_2 equal to $a'o_1$ and bB_2 equal to $b'o_2$, on its opposite sides (as a' and b' are on opposite sides of xy). Join A_2 with B_2 cutting ab at the H.T. of the line.

Note that $A_1B_1 = A_2B_2 = AB$ and that θ and ϕ are the inclinations of AB with the H.P. and the V.P. respectively.



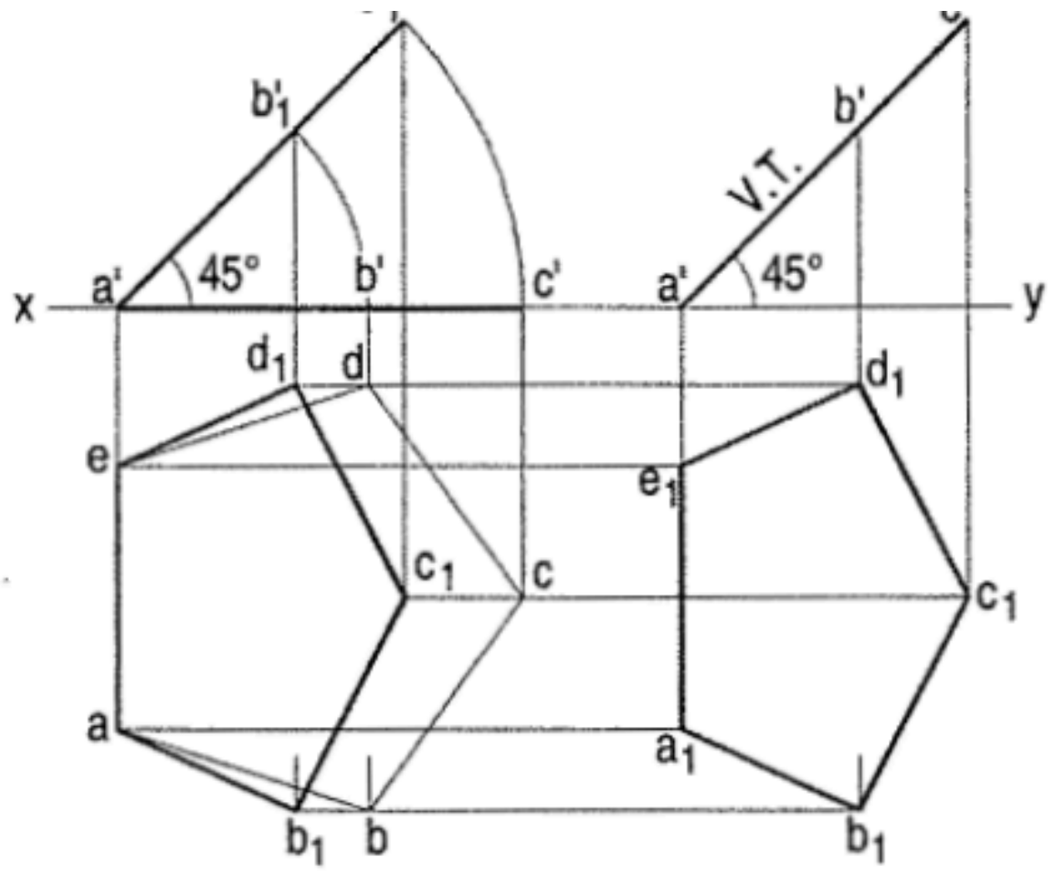
Q4. A regular pentagon of 25 mm side has one side on the ground. Its plane is inclined at 45 deg to the H.P. and perpendicular to the V.P. Draw its projections and show its traces.

Assuming it to be parallel to the H.P.

(i) Draw the pentagon in the top view with one side perpendicular to xy Project the front view. It will be the line $a'c'$ contained by xy.

(ii) Tilt the front view about the point a', so that it makes 45° angle with xy.

(iii) Project the new top view $ab_1c_1d_1e$ upwards from this front view and horizontally from the first top view. It will be more convenient if the front view is reproduced in the new position separately and the top view projected from it, as shown in . The V.T. coincides with the front view and the H.T. is perpendicular to xy, through the point of intersection between xy and the front view-produced.

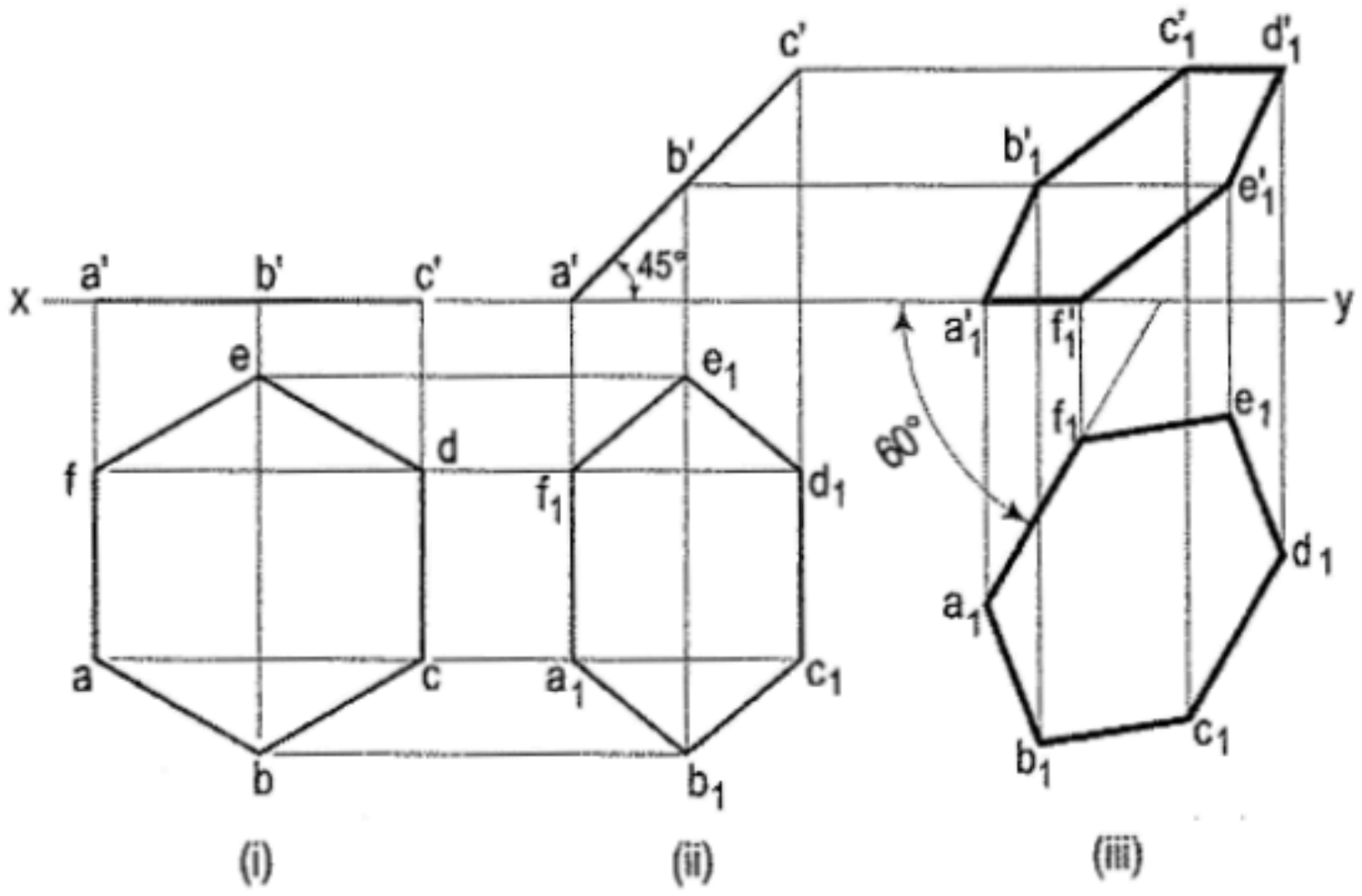


Q5. Draw the projections of a regular hexagon of 25 mm side, having one of its sides in the H.P. and inclined at 60 deg to the V.P., and its surface making an angle of 45° with the H.P.

(i) Draw the hexagon in the top view with one side perpendicular to *xy*. Project the front view *a'c'* in *xy*.

(ii) Draw *a'c'* inclined at 45° to *xy* keeping *a'* or *c'* in *xy* and project the second top view.

(iii) Reproduce this top view making a_1f_1 inclined at 60° to *xy* and project the final front view.



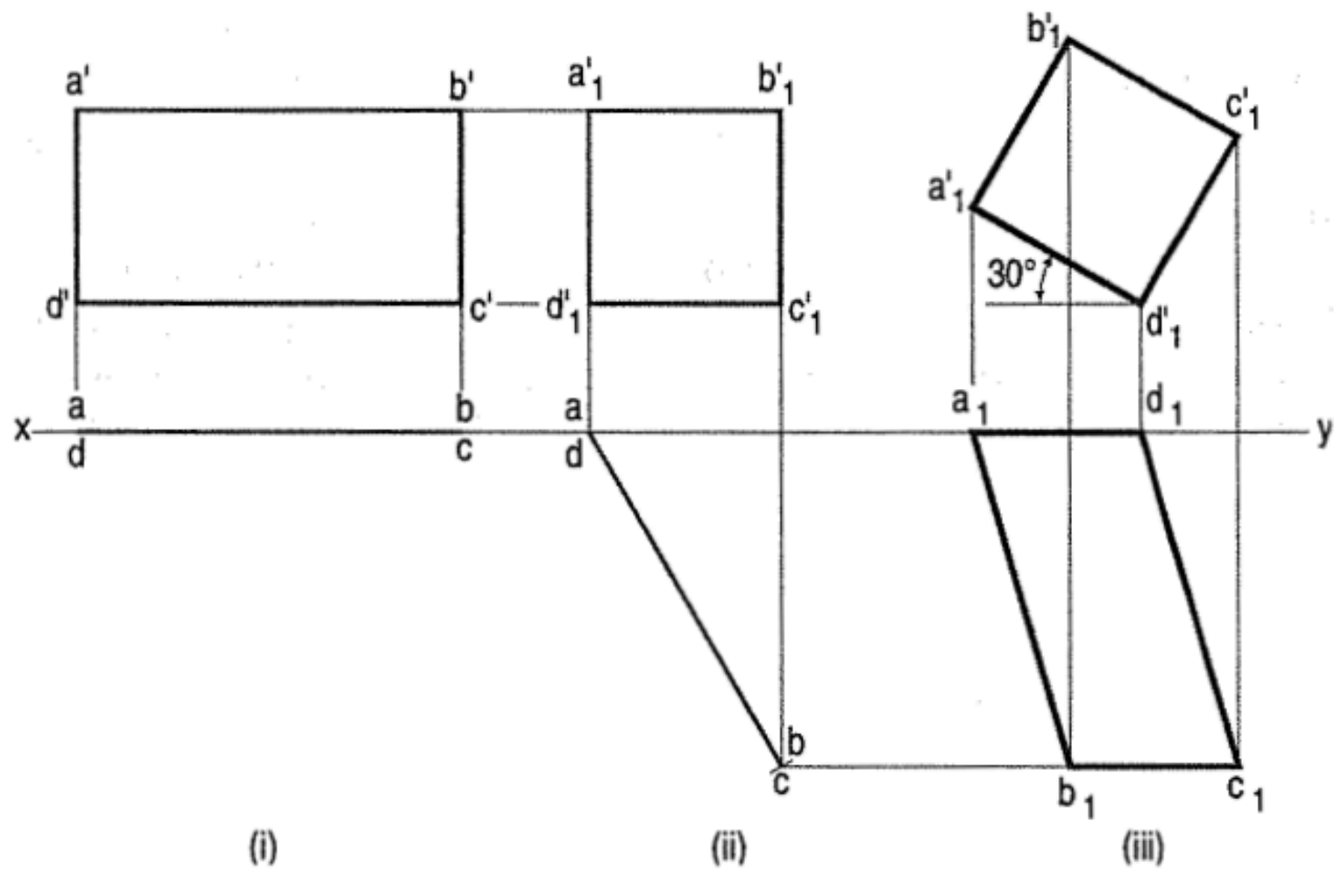
Q6. A thin rectangular plate of sides 60 mm x 30 mm has its shorter side in the V.P. and inclined at 30° to the H.P. Project its top view if its front view is a square of 30 mm long sides.

As the front view of the plate is a square, its surface must be inclined to the V.P. Hence, assume the plate to be in the V.P. with its shorter edge perpendicular to the H.P.

(i) Draw the front view *a'b'c'd'* and project the top view *ab* in *xy*

(ii) The line *ab* should be so inclined to *xy* that the front view becomes a square. Therefore, draw the square $a'_1b'_1c'_1d'_1$ of side equal to *a'd'*. With *a* as centre and radius equal to *ab* draw an arc cutting the projector through b'_1 at *b*. Then *ab* is the new top view.

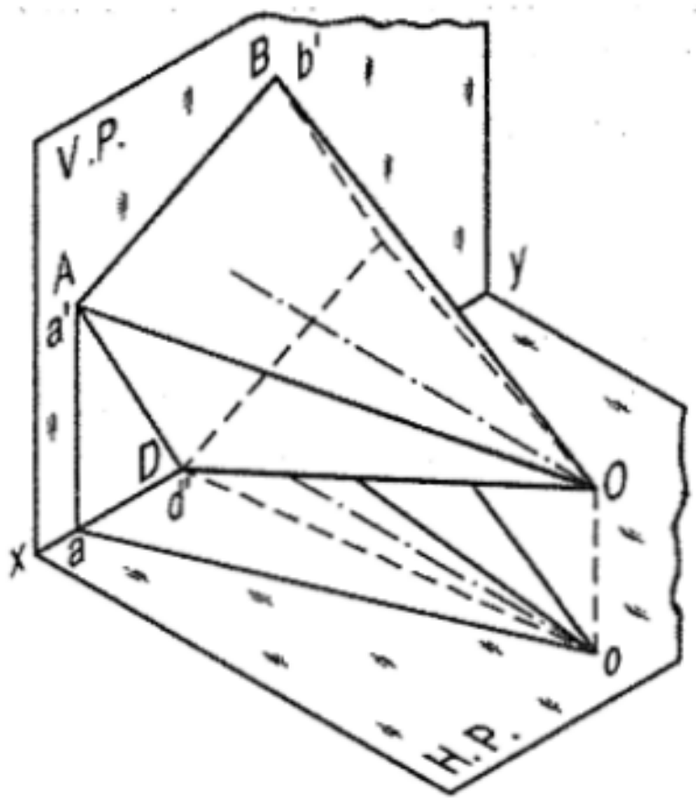
(iii) Reproduce the second front view in such a way that $a'_1d'_1$ makes 30° angle with *xy*. Project the final top view as shown.

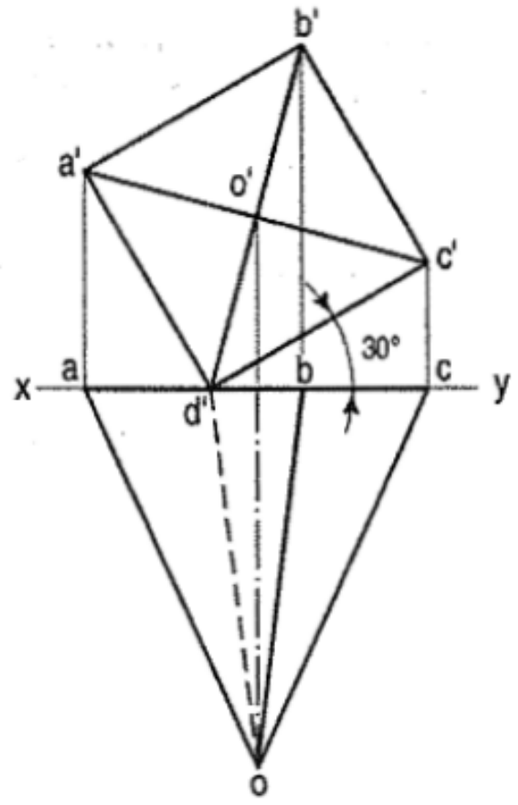


Q1. A square pyramid, base 40 mm side and axis 65 mm long, has its base in the V.P. One edge of the base is inclined at 30° to the H.P. and a corner contained by that edge is on the H.P. Draw its projections.

(i) Draw a square in the front view with the corner d' in xy and the side $d'c'$ inclined at 30° to it. Locate the centre o' and join it with the corners of the square.

(ii) Project down all the corners in xy (because the base is in the V.P.). Mark the apex o on a projector through o'. Draw lines for the slant edges and complete the top view.





Q2. Draw the projections of a pentagonal prism, base 25 mm side and axis 50 mm long, resting on one of its rectangular faces on the H.P., with the axis inclined at 45° to the V.P.

In the simple position, assume the prism to be on one of its faces on the ground with the axis perpendicular to the V.P.

Draw the pentagon in the front view with one side in xy and project the top view [fig. 13-21(i)].

The shape and size of the figure in the top view will not change, so long as the prism has its face on the H.P. The respective distances of all the corners in the front view from xy will also remain constant.

Method I: [fig. 13-21(ii)]**:**

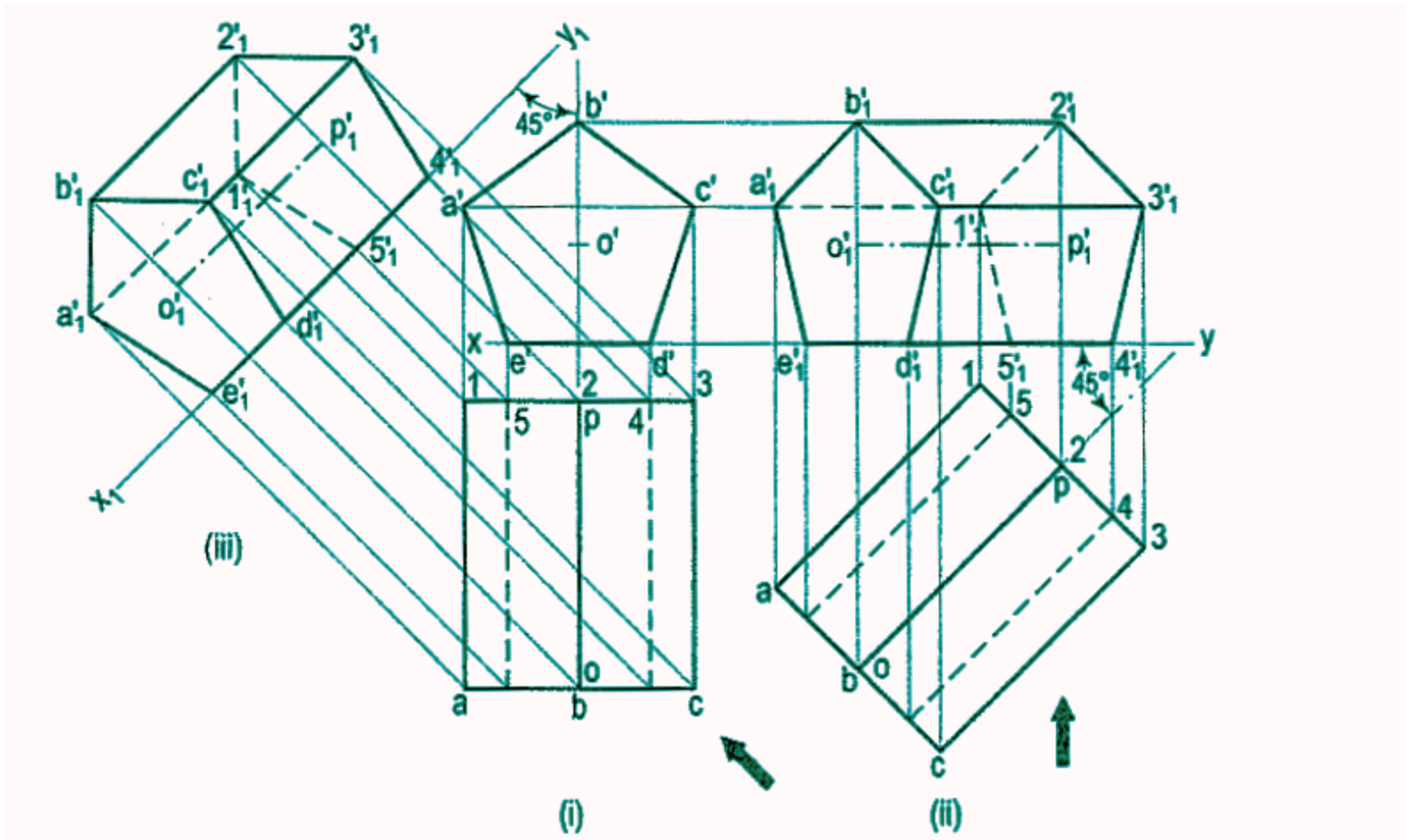
(i) Alter the position of the top view, i.e. reproduce it so that the axis is inclined at 45° to xy. Project all the points upwards from this top view and horizontally from the first front view, e.g. a vertical from a intersecting a horizontal from a' at a point a'_1.

(ii) Complete the pentagon $a'_1b'_1c'_1d'_1e'_1$ for the fully visible end of the prism. Next, draw the lines for the longer edges and finally, draw the lines for the edges of the other end. Note carefully that the lines $a'_11'_1$, $1'_12'_1$ and $1'_15'_1$ are dashed lines. $e'_15'_1$ is also hidden but it coincides with other visible lines.

Method II: [fig. 13-21(iii)]**:**

(i) Draw a new reference line x_1y_1, making 45° angle with the top view of the axis, to represent an auxiliary vertical plane.

(ii) Draw projectors from all the points in the top view perpendicular to x_1y_1 and on them, mark points keeping the distance of each point from x_1y_1 equal to its distance from xy in the front view. Join the points as already explained. The auxiliary front view and the top view are the required projections.



Q3. A hexagonal pyramid, base 25 mm side and axis 50 mm long, has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the V.P. Draw its projections.

In the initial position assume the axis to be perpendicular to the H.P.

Draw the projections with the base in *xy* and its one edge perpendicular to the V.P.

If the pyramid is now tilted about the edge *AF* (or *CD*) the axis will become inclined to the H.P. but will remain parallel to the V.P. The distances of all the corners from the V.P. will remain constant.

The front view will not be affected except in its position in relation to *xy*. The new top view will have its corners at same distances from *xy*, as before.

Method I:

(i) Reproduce the front view so that the axis makes 30° angle with *xy* and the point *a'* remains in *xy*.

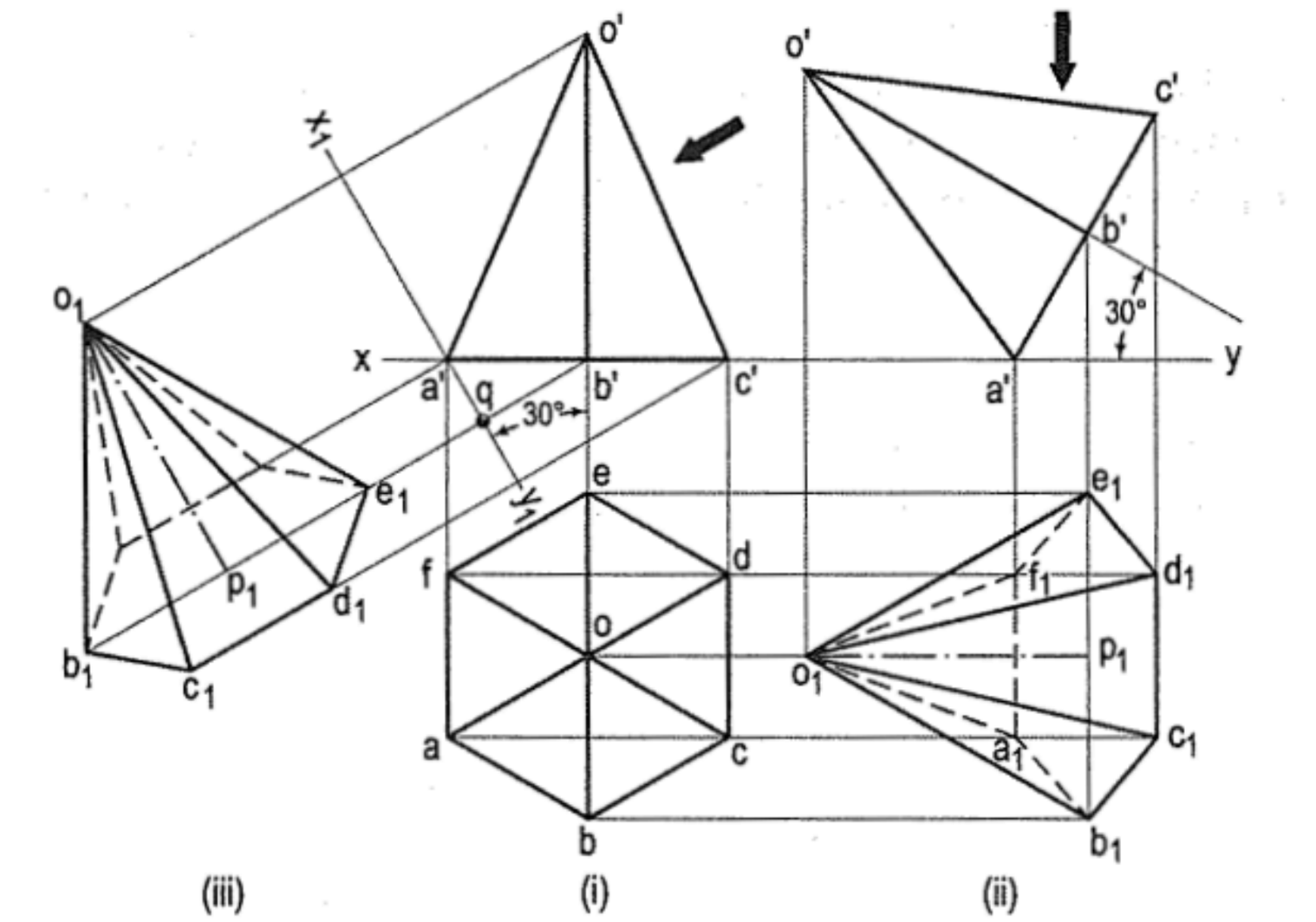
(ii) Project all the points vertically from this front view and horizontally from the first top view. Complete the new top view by drawing (a) lines joining the apex o'_1 with the corners of the base and (b) lines for the edges of the base.

The base will be partly hidden as shown by dashed line a_1b_1, e_1f_1 and f_1a_1. Similarly o_1f_1 and o_1a_1 are also dashed lines.

Method II:

(i) Through *a'* draw a new reference line x_1y_1 inclined at 30° to the axis, to represent an auxiliary inclined plane.

(ii) From the front view project the required top view on x_1y_1, keeping the distance of each point from x_1y_1 equal to the distance of its first top view from *xy*, viz. $e_1q = eb'$ etc.



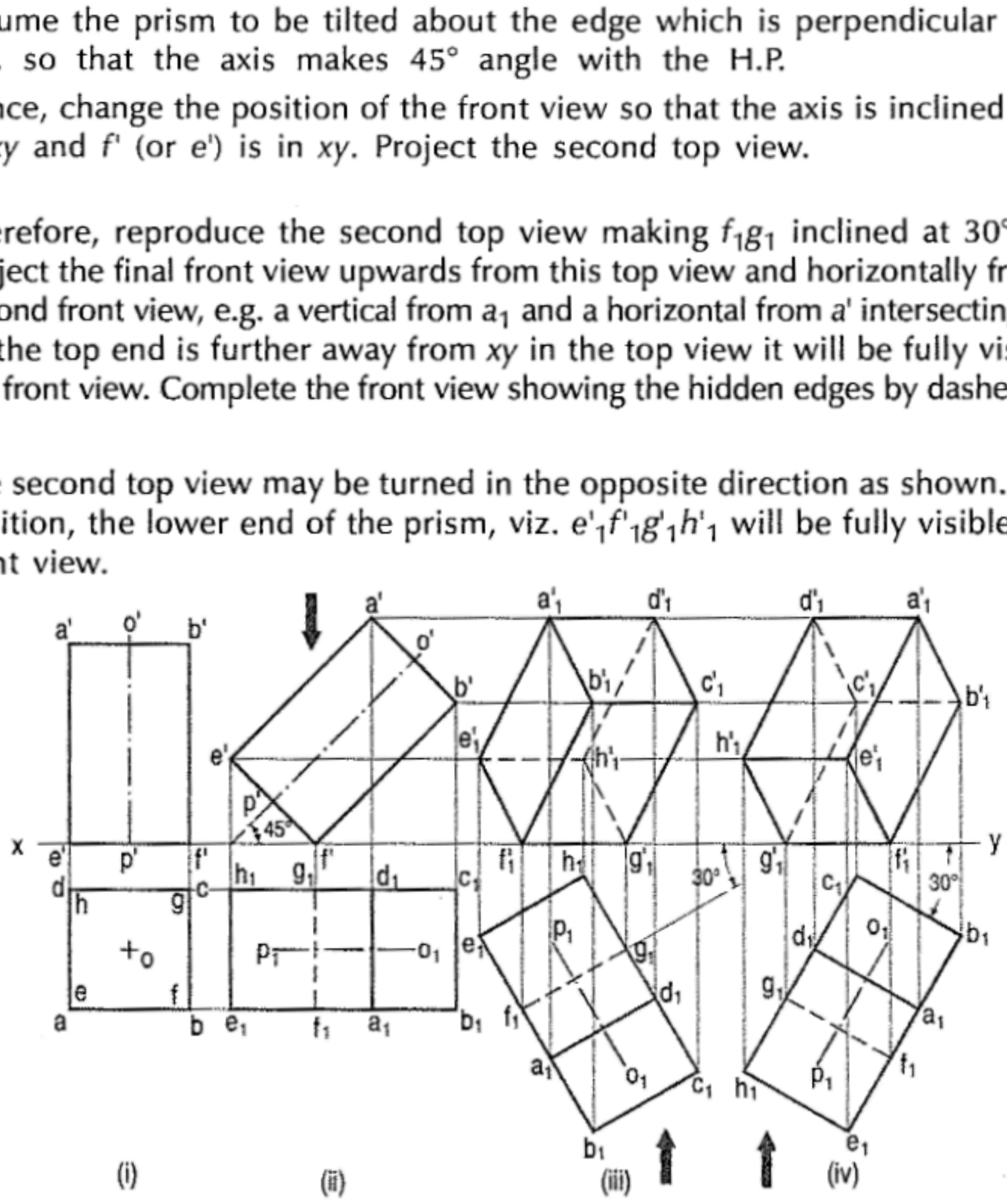
Q4. A square prism, base 40 mm side and height 65 mm, has its axis inclined at 45° to the H.P. and has an edge of its base, on the H.P and inclined at 30° to the V.P. Draw its projections.

(i) Assuming the prism to be resting on its base on the ground with an edge of the base perpendicular to the V.P., draw its projections.
Assume the prism to be tilted about the edge which is perpendicular to the V.P., so that the axis makes 45° angle with the H.P.

(ii) Hence, change the position of the front view so that the axis is inclined at 45° to *xy* and f' (or e') is in *xy*. Project the second top view.

(iii) Therefore, reproduce the second top view making f_1g_1 inclined at 30° to *xy*. Project the final front view upwards from this top view and horizontally from the second front view, e.g. a vertical from a_1 and a horizontal from a' intersecting at a'_1. As the top end is further away from *xy* in the top view it will be fully visible in the front view. Complete the front view showing the hidden edges by dashed lines.

(iv) The second top view may be turned in the opposite direction as shown. In this position, the lower end of the prism, viz. $e'_1f'_1g'_1h'_1$ will be fully visible in the front view.

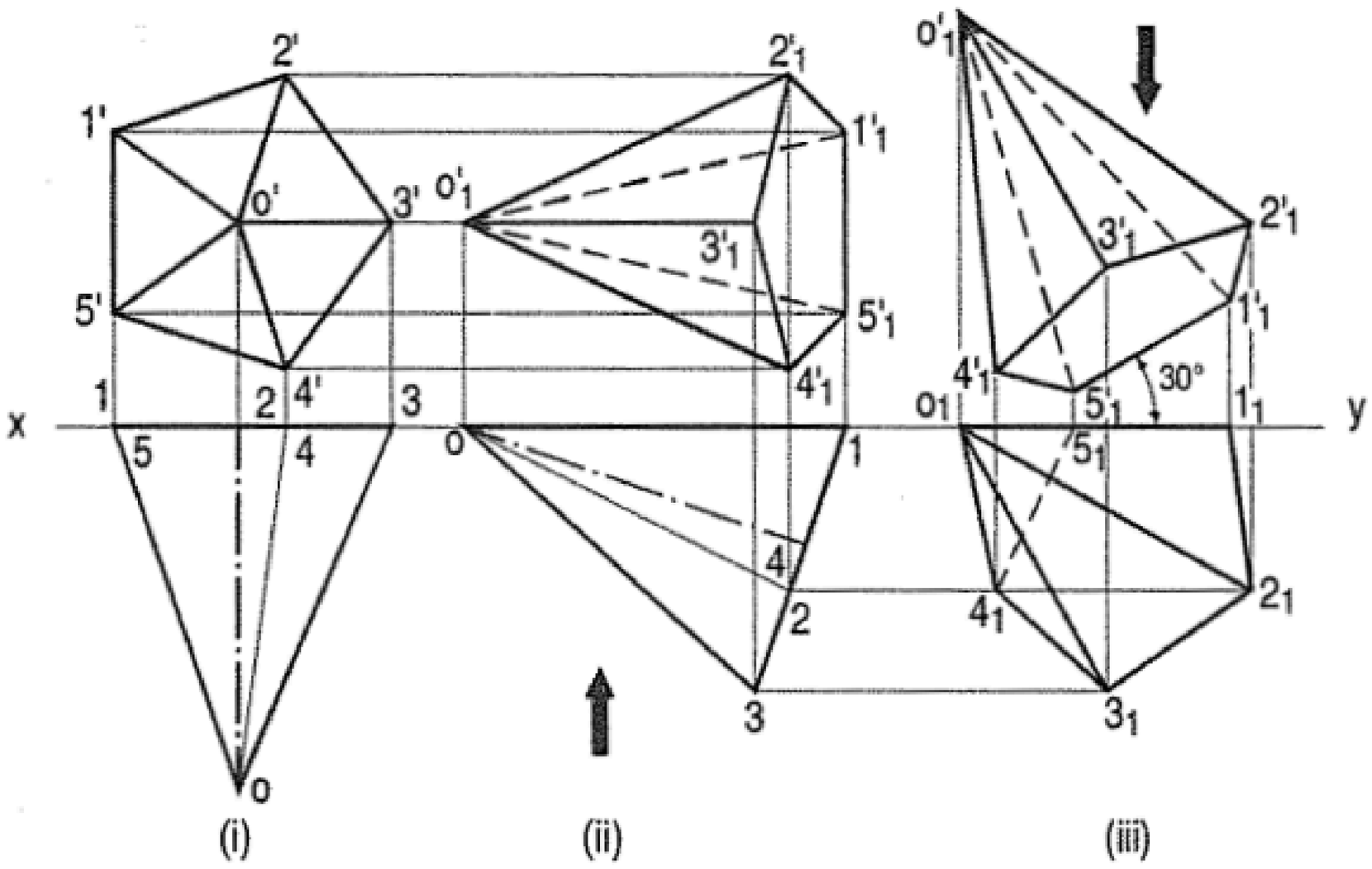


Q5. A pentagonal pyramid, base 25 mm side and axis 50 mm long has one of its triangular faces in the V.P. and the edge of the base contained by that face makes an angle of 30° with the H.P. Draw its projections

(i) In the initial position, assume the pyramid as having its base in the V.P. and an edge of the base perpendicular to the H.P. The front view will have to be drawn first and the top view projected from it.

(ii) Change the position of the top view so that the line o1 (for the face o1 5) is in *xy*. Project the second front view.

(iii) Tilt this front view so that the line $1'_1 5'_1$ makes 30° angle with *xy*. Project the final top view. Note that the base is not visible in the top view as it is nearer *xy* in the front view.

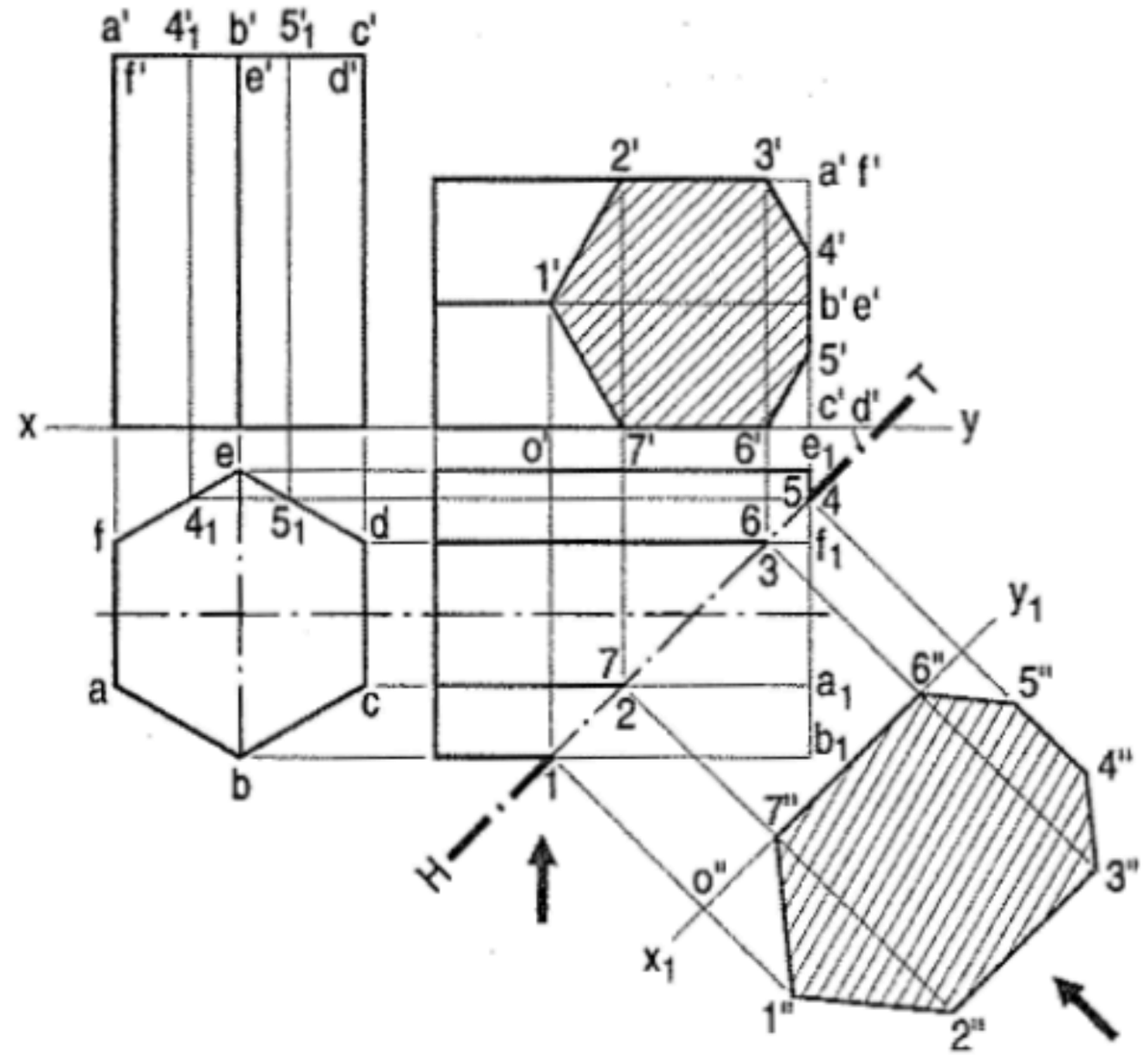


Q1. A hexagonal prism, has a face on the H.P. and the axis parallel to the V.P. It is cut by a vertical section plane, the H.T. of which makes an angle of 45° with xy and which cuts the axis at a point 20 mm from one of its ends. Draw its sectional front view and the true shape of the section. Side of base 25 mm long; height 65 mm.

(i) Draw the front view and the top view of the prism and show the H.T. of the section plane in the top view. Name in proper sequence, the points at which the lines are cut.

(ii) Project them on the corresponding lines in the front view. The positions of points 4 and 5 cannot be located directly. Hence, project them on the first top view to 4_1 on *ef* and 5_1 on *ed*. From this top view, obtain their positions $4'_1$ and $5'_1$ on the corresponding lines in the first front view. As the two front views are identical, these points can now be transferred to the second front view by making *e*'4' equal to *e*'$4'_1$ and *e*'5' equal to *e*'$5'_1$. 4' and 5' are the projections of points 4 and 5 respectively. Complete the sectional front view as shown.

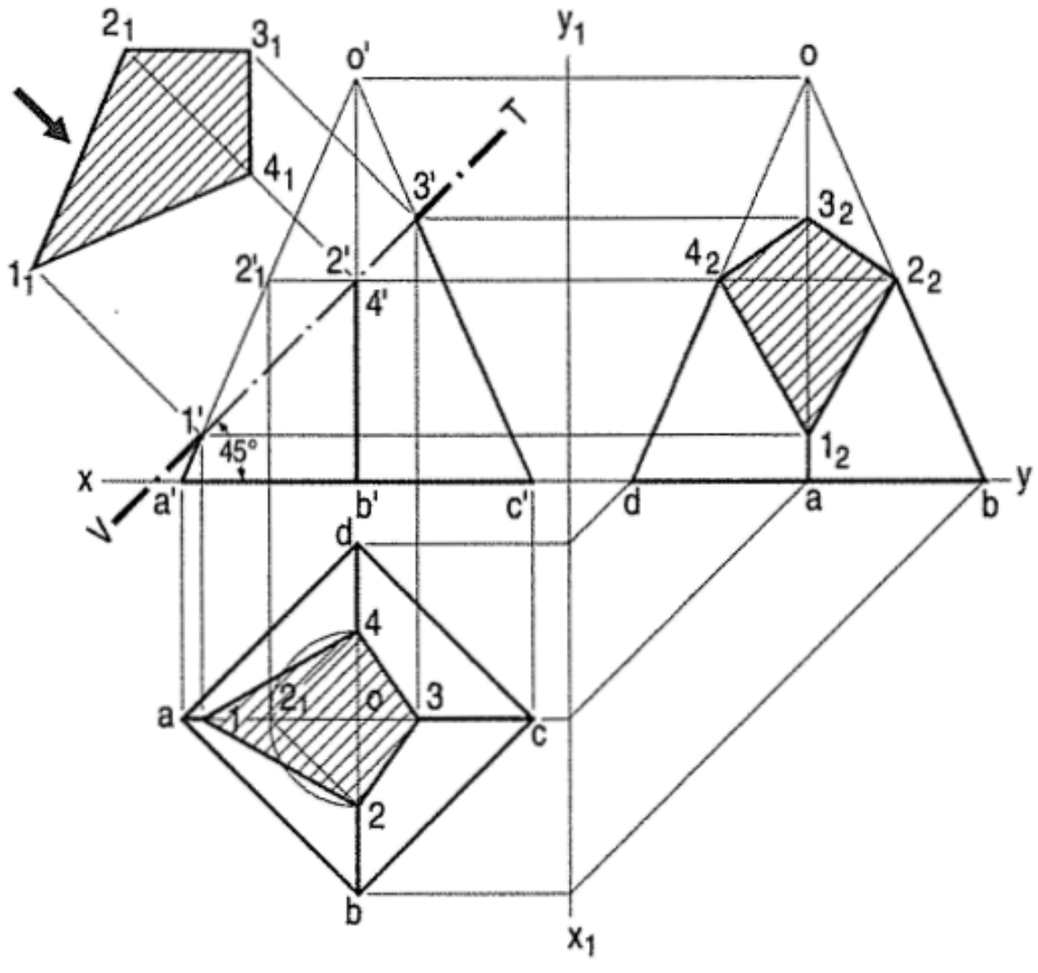
(iii) Obtain the true shape of the section on x_1y_1 as explained in problem 14-3, making *o*"1" equal to *o*'1', etc.



Q2. A triangular pyramid, having base 40 mm side and axis 50 mm long, is lying on the H.P. on one of its faces, with the axis parallel to the V.P. A section plane, parallel to the V.P. cuts the pyramid at a distance of 6 mm from the axis. Draw its sectional front view and the top view.

(i) Draw the projections of the pyramid in the required position and show a line H.T. (for the cutting plane) in the top view parallel to *xy* and 6 mm from the axis.

(ii) Project points 1, 2 and 3 (at which the edges are cut) on corresponding edges in the front view and join them. Figure 1' 2' 3' shows the true shape of the section.

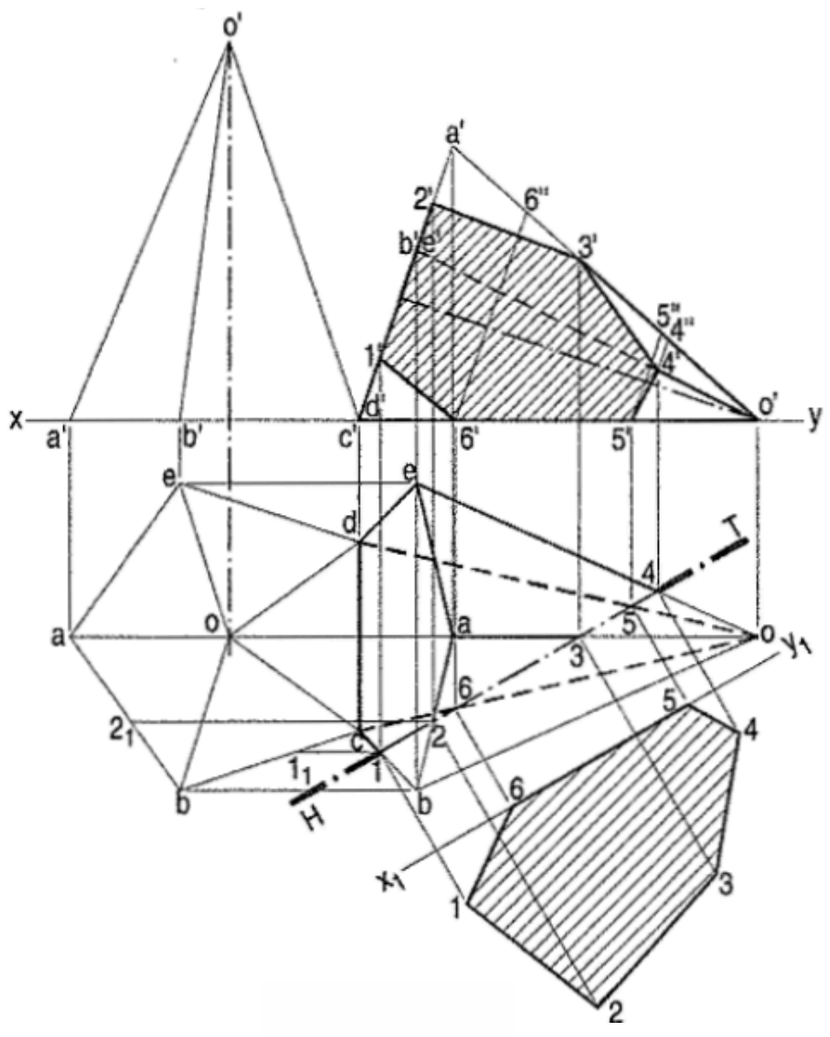


Q3. A pentagonal pyramid, base 30 mm side and axis 60 mm long, is lying on one of its triangular faces on the H.P. with the axis parallel to the V.P. A vertical section plane, whose H.T. bisects the top view of the axis and makes an angle of 30° with the reference line, cuts the pyramid, removing its top part. Draw the top view, sectional front view, true shape of the section and development of the surface of the remaining portion of the pyramid.

(i) Draw the projections of the pyramid in the required position and show a line H.T. (for the cutting plane) in the top view parallel to *xy* and 6 mm from the axis.

(ii) Project points 1, 2 and 3 (at which the edges are cut) on corresponding edges in the front view and join them. Figure 1' 2' 3' shows the true shape of the section.

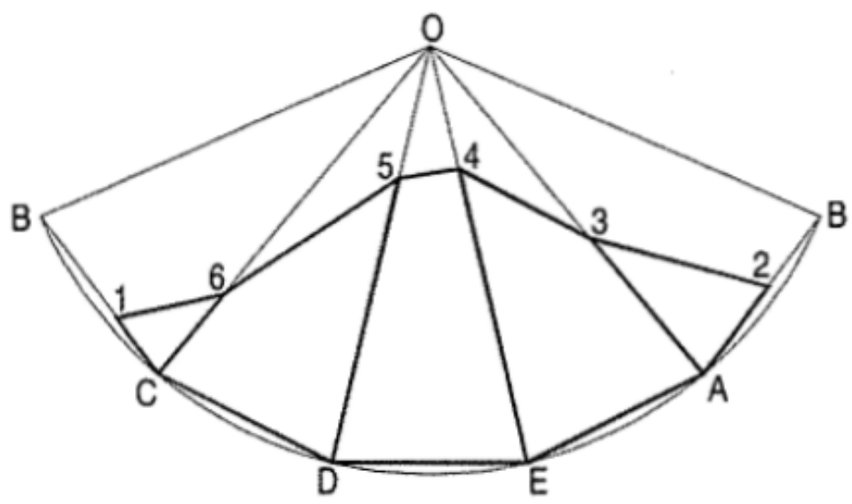
(iii) Obtain the true shape of the section on x_1y_1 as a new reference line drawn parallel to the H.T.



Development

(i) With any point *O* as centre and radius *o'a'*, draw an arc and construct the development of the whole pyramid. Mark points on it, taking the positions of 1 and 2 from the first top view and those of other points by projecting them on the true-length-line *o'a'*.

(ii) Draw lines joining these points and complete the development as shown in the figure.



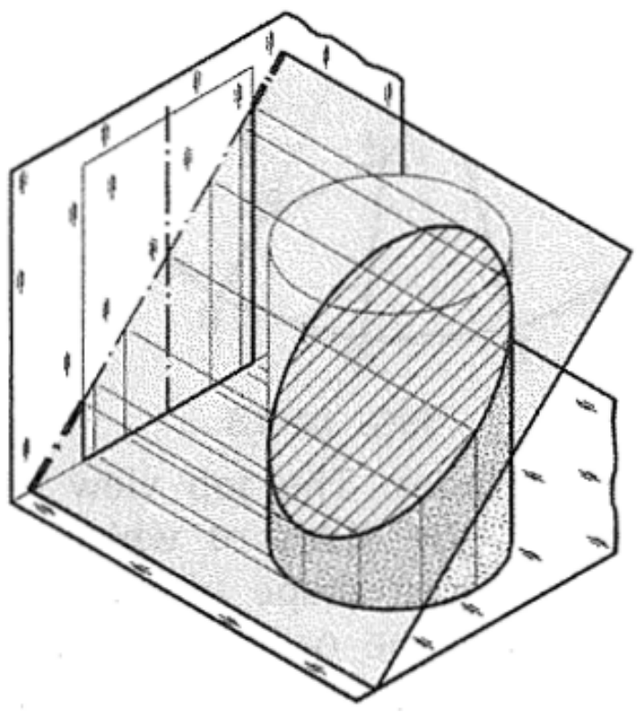
Q4. A cylinder of 40 mm diameter, 60 mm height and having its axis vertical, is cut by a section plane, perpendicular to the V.P., inclined at 45 deg to the H.P. and intersecting the axis 32 mm above the base. Draw its front view, sectional top view, sectional side view and true shape of the section.

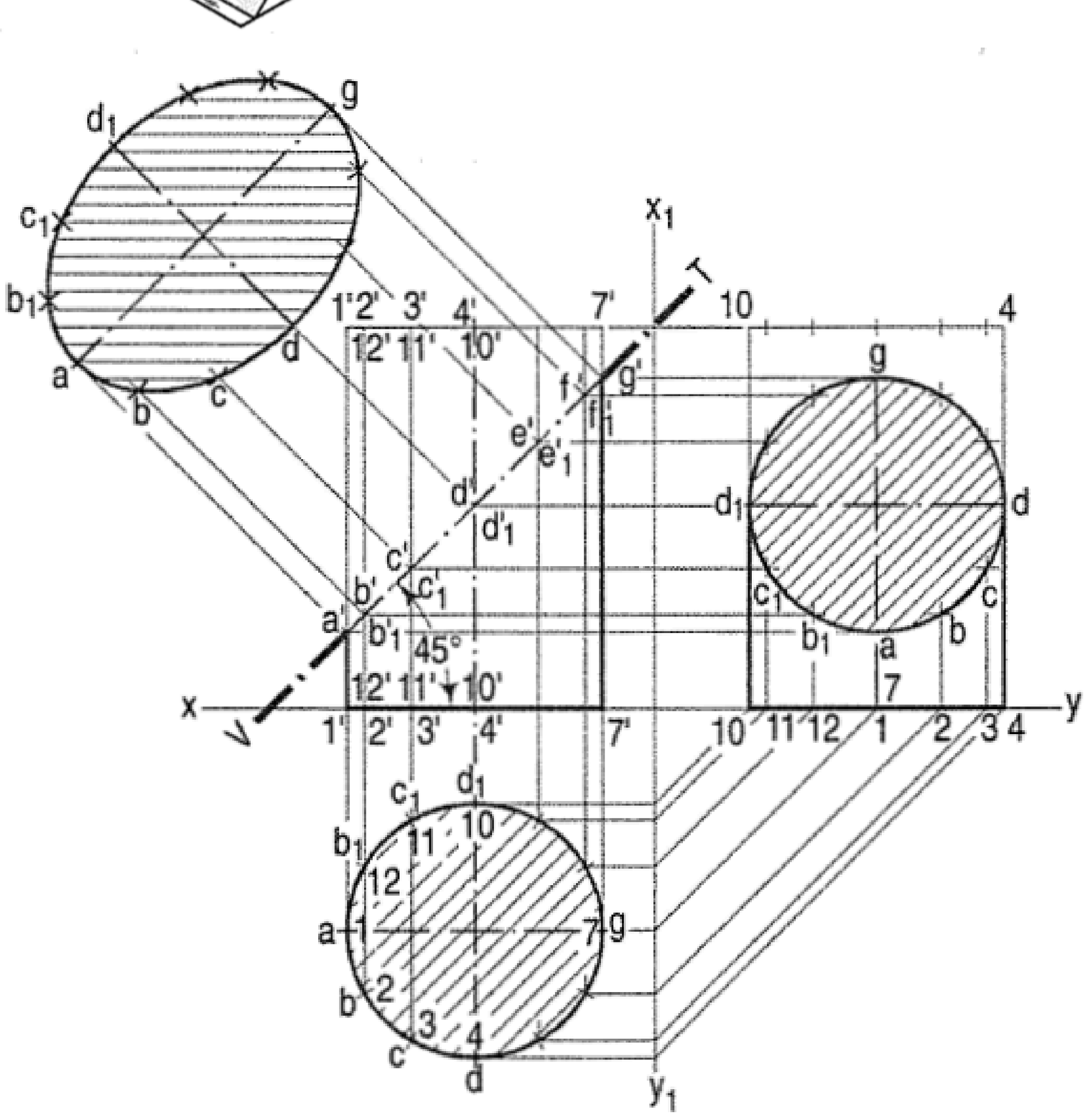
As the cylinder has no edges, a number of lines representing the generators may be assumed on its curved surface by dividing the base-circle into, say 12 equal parts.

(i) Name the points at which these lines are cut by the V.T. In the top view, these points lie on the circle and hence, the same circle is the top view of the section. The width of the section at any point, say c', will be equal to the length of the chord cc_1 in the top view.

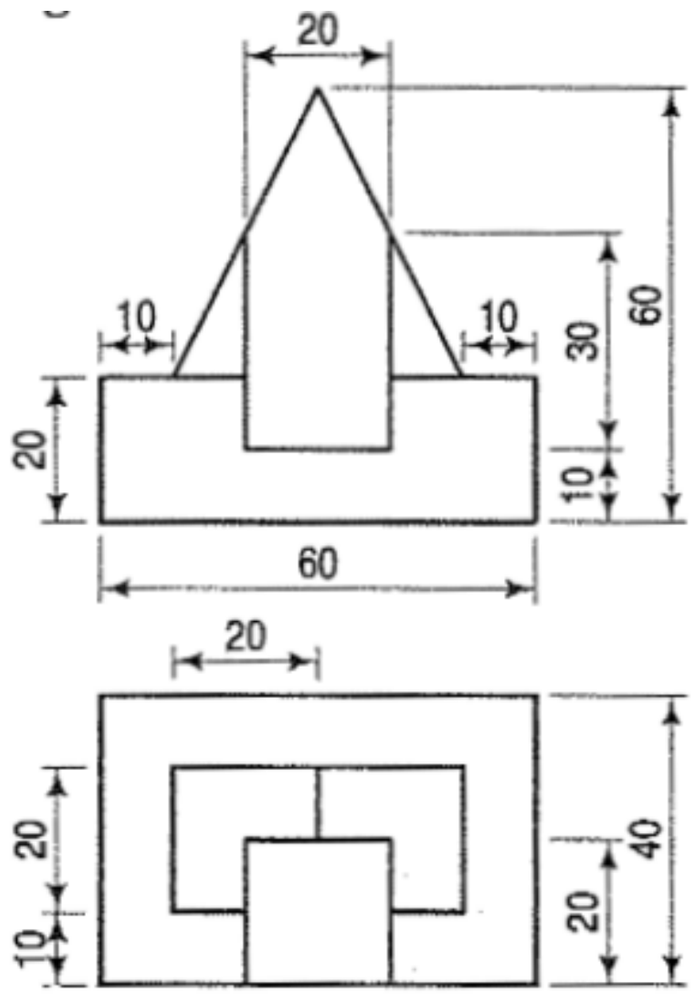
(ii) The true shape of the section may be drawn around the centre line *ag* drawn parallel to V.T. as shown. It is an ellipse the major axis of which is equal to the length of the section plane viz. *a'g'*, and the minor axis equal to the diameter of the cylinder viz. dd_1.

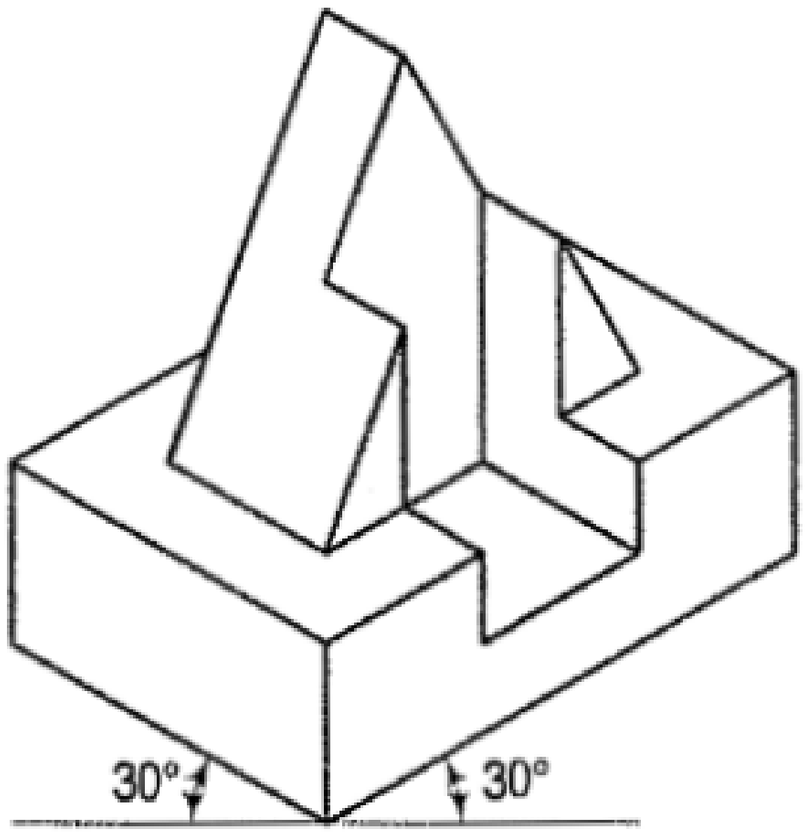
(iii) Project the sectional side view as shown. The section will be seen as a circle because the section plane makes 45° angle with *xy*.



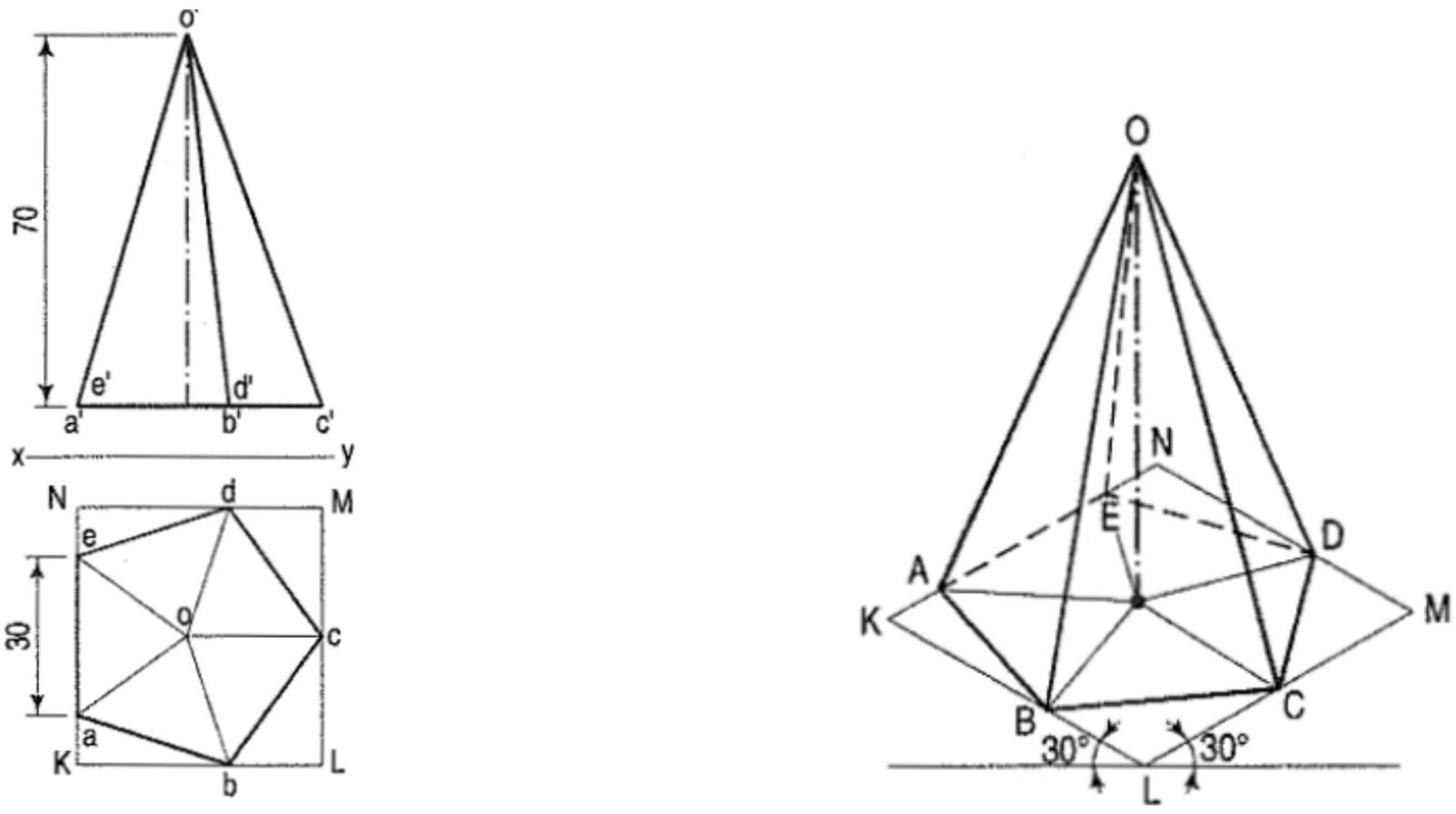


Q1. The orthographic projections of the object is shown in fig. Draw the isometric view of the object.

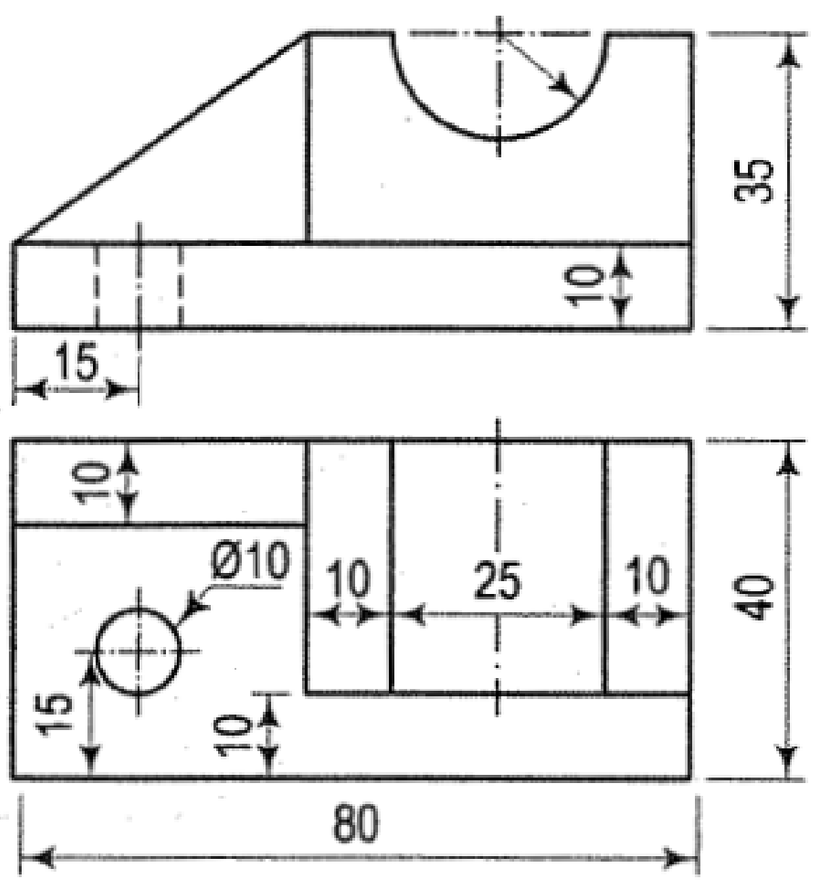


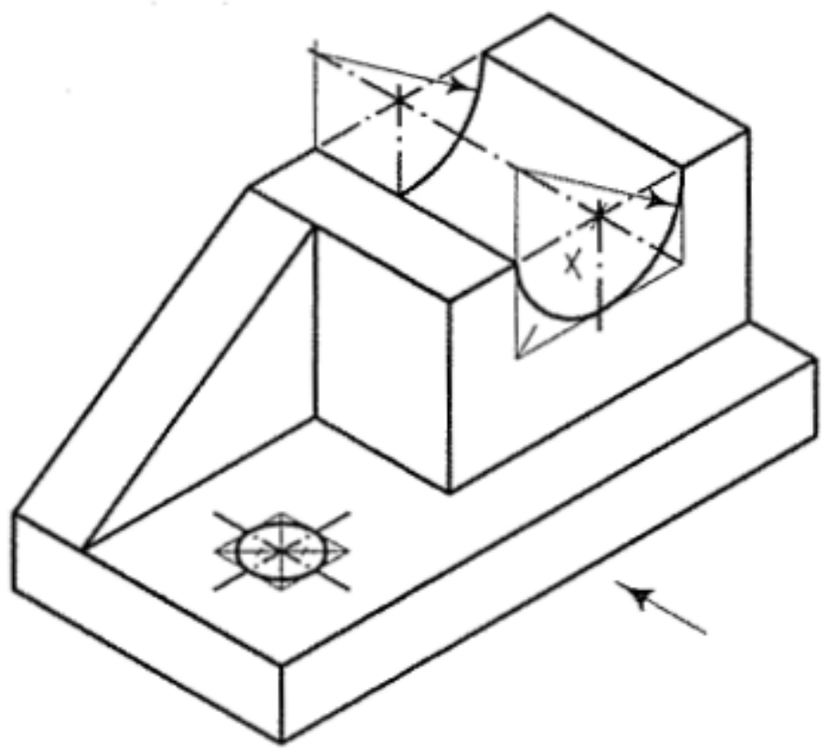


Q2. The projection of pentagonal pyramid is shown in fig. Draw its isometric view.



Q3. Draw the isometric view of the casting shown in two views in fig.





Q4. A pictorial view of a machine bracket is shown in the figure. Draw using the first angle projection method front view, top view and right end side view.

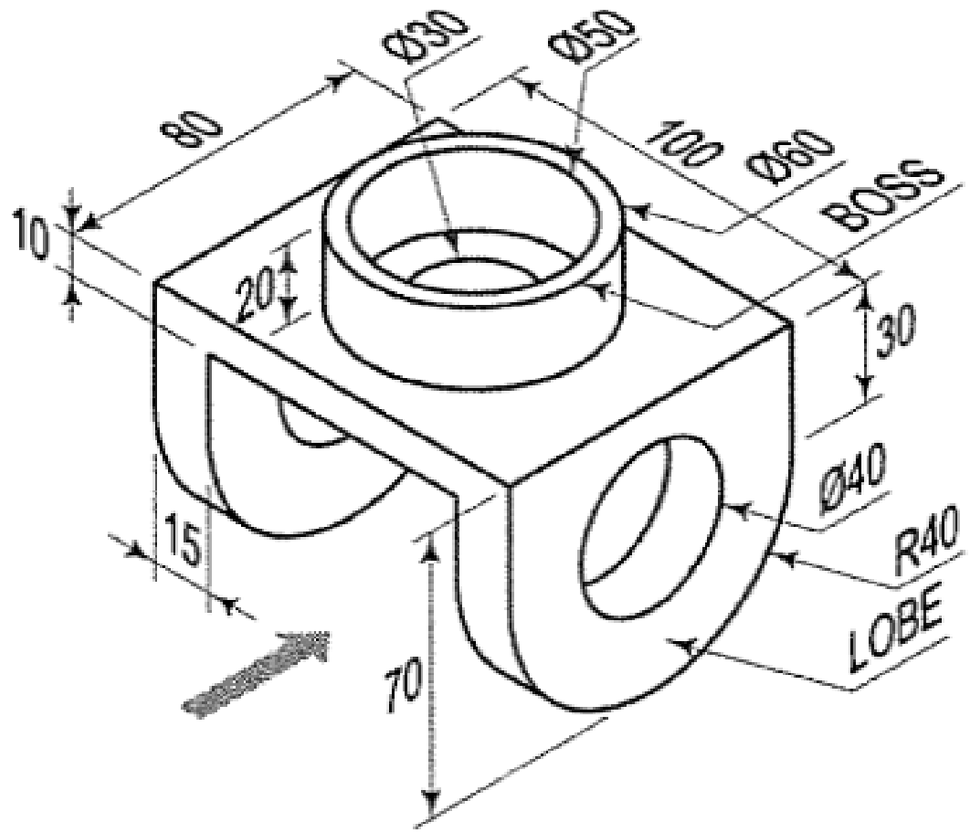
Assume that you are viewing the object in the direction of the arrow towards the imaginary V.P. What will you see? It is a rectangle of height 70 mm and width 100 mm on V.P. This will be the front view. To draw this view:

(i) Draw a Reference line *xy*, which represents the intersecting line of the planes V.P. and H.P.

(ii) Draw rectangular block of size 100 × 70 and the thickness is 15 mm each on the two parallel portions of the slot.

(iii) Now on the top of this rectangular block draw a rectangle of size 60 mm × 20 mm. This rectangle represents a boss of ø60 on the rectangular block.

(iv) Draw a hole of ø50 in the above boss. Note carefully that the lines are dashed lines as they are invisible from the front side. The ø30 hole is in the rectangular block.



(v) Draw the two holes on the parallel edges of the rectangular block. These holes are represented by dotted lines in the front view as they are hidden. This completes the front view construction.

(vi) To draw the top view, project all the details from front view.

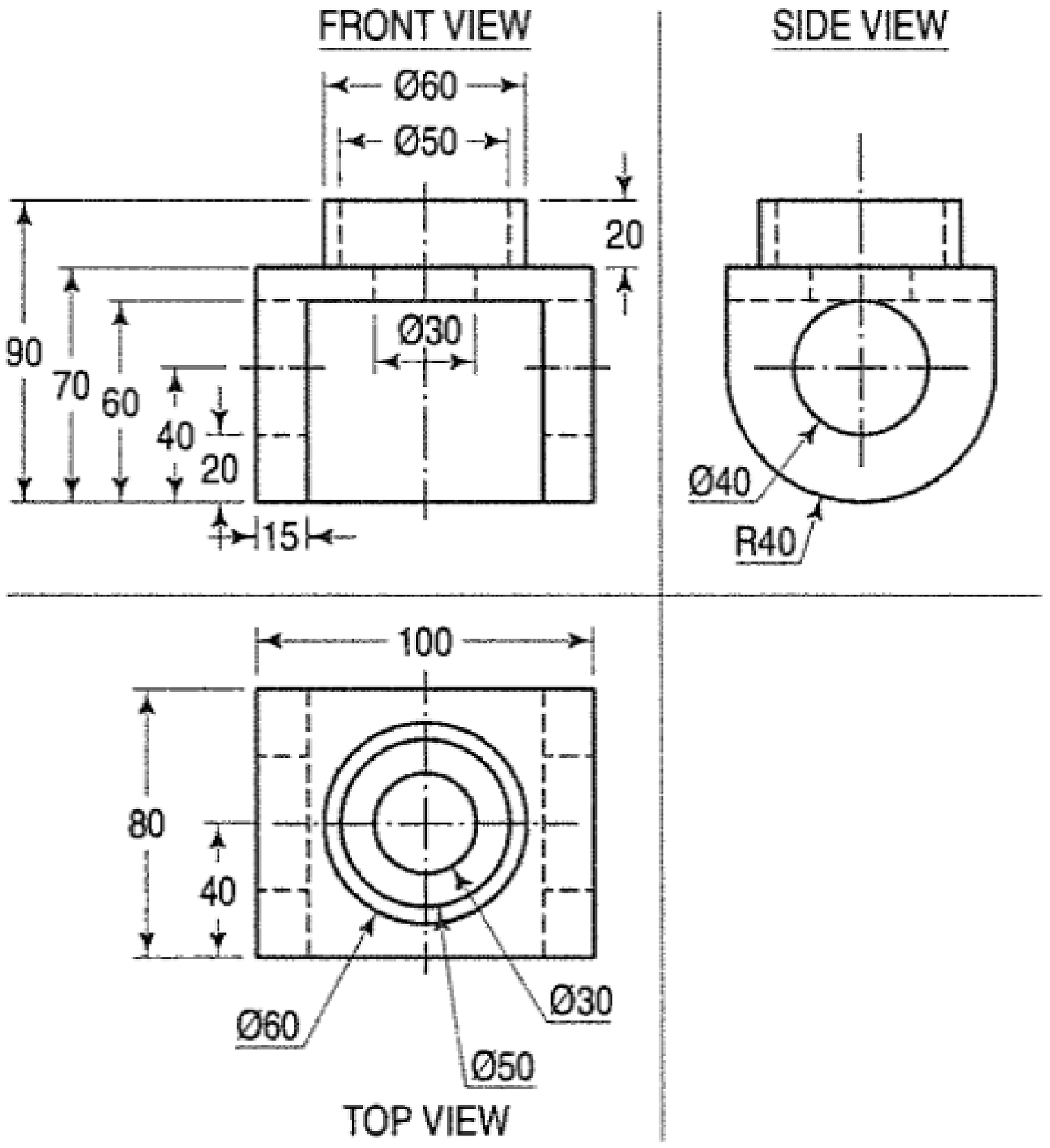
(vii) Draw a vertical centre line form the centre of the rectangular block.

(viii) The top view of the rectangular block appears as a rectangle of size 100 × 80.

(ix) Draw the holes in the centre of the rectangle. These holes are drawn taking the common centre point on vertical centre line.

(x) The two holes of ø40 on the lobes are not visible, so they are projected as dashed lines. This completes the top view.

(xi) To draw the side view, draw projectors from all the points in front view and top view to side view. Join the intersection points. This completes the side view.



NOTE- YOU SHOULD FOCUS ON CONCEPT OF THESE QUESTIONS. IN THE EXAM QUESTION DATA MAY CHANGE BUT, CONCEPT WILL BE SAME.